Arista Access Node WFD-ACC-71 Hardware/Environmental Fault Resolution SOP

Node: WFD-ACC-71 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6541

Description

The described failure domain of Hardware/Environmental Fault on the Arista Access Node WFD-ACC-71 is primarily attributed to the inherent susceptibility of ASIC memory parity errors and line card crashes, precipitated by power supply redundancy loss. The Arista Access Node, being a critical component of the Whitefield POP infrastructure, relies heavily on its ASIC (Application-Specific Integrated Circuit) for handling data-plane operations. The memory parity checks within the ASIC are designed to detect and correct data corruption; however, a parity error can lead to a line card crash if left unchecked. Furthermore, a loss of power supply redundancy can exacerbate the situation, rendering the node inoperable. This failure domain is further compounded by the specific architectural design of the Arista Access Node, which necessitates a swift and precise resolution to prevent widespread network instability.

Pre-requisite Checks

- id: PHYS-01, description: Verify that the remote hands are dispatched to the Whitefield POP to facilitate physical access to the affected node.

- id: ENV-01, description: Check the redundant link status to ensure that it is operational and can be utilized for network redundancy purposes.

- id: ENV-02, description: Verify that the environmental conditions at the Whitefield POP are within the recommended operating parameters for the Arista Access Node.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash due to power supply redundancy loss	Medium
Inadequate physical or environmental checks prior to deployment	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware internal, Expected Output: ASIC memory parity error detected, Description: This command helps to diagnose the ASIC memory parity error issue by providing a detailed view of the internal hardware state.

Command: show system redundancy, Expected Output: Redundant power supply lost, Description: This command helps to diagnose the power supply redundancy loss issue by providing a detailed view of the system redundancy state.
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```
Command: show interface, Expected Output: Line card crash detected, Description: This command helps to diagnose the line card crash issue by providing a detailed view of the interface status.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: hw-module replace asic, Description: This command helps to replace the faulty ASIC with a new one to resolve the memory parity error issue.

Step 2: Step: Reboot the node, Command: system reboot, Description: This command helps to reboot the node to resolve any potential software issues or to reinitialize the hardware.

Step 3: Step: Verify the redundant link status, Command: show system redundancy, Description: This command helps to verify the redundant link status to ensure that it is operational and can be utilized for network redundancy purposes.

Post-Check Verification

```
Command: show hardware internal, Expected Output: ASIC memory parity error resolved, Description: This command helps to verify that the ASIC memory parity error issue is resolved.

Command: show system redundancy, Expected Output: Redundant power supply restored, Description: This command helps to verify that the redundant power supply is restored.

Command: show interface, Expected Output: Line card crash resolved, Description: This command helps to verify that the line card crash issue is resolved.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: hw-module restore asic, Description: This command helps to restore the original ASIC in case the replacement ASIC does not resolve the issue.

- Step: Reboot the node, Command: system reboot, Description: This command helps to reboot the node to restore the original state.